

CSQ413 Seminar Abstract

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Topic 1 - Generative Agents in Agent - Based Modelling: Overview, Validation, and Emerging Challenges

Base Paper Details

Authors: Carlo Adornetto, Adrian Mora, Kai Hu, Leticia Izquierdo Garcia, Parfait Atchade-Adelomou, Gianluigi Greco, Luis Alberto Alonso Pastor, Kent Larson

Year: 2025

Journal: IEEE Transactions on Artificial Intelligence

Publisher: IEEE

DOI: 10.1109/TAI.2025.3566362

Abstract

The integration of Large Language Model (LLM)-driven generative agents (GAs) has emerged as a disruptive frontier redefining traditional computational social science and agent-based modeling (ABM). Moving beyond classical mathematical frameworks and rigid, predefined rationality assumptions, this new paradigm establishes conversational, adaptive software agents that simulate believable human behaviors, dynamic memory streams, and subjective emotional states within open-domain configurations. This seminar explores the operational mechanics of generative agent-based models (GABMs), deconstructing core architectural implementations across peer-to-peer social influence networks and urban community simulators. The discussion critically maps the structural validation landscape, balancing empirical data matching with advanced microface and macroface human-centric verification workflows. Furthermore, the seminar addresses prominent engineering hurdles, including token hallucinations, train-test contamination loops, and the high computational costs associated with scaling multi-agent environments. Based on a comprehensive review, this topic provides the necessary theoretical depth and practical methodologies to analyze the future of cognitively driven agentic simulations.

Topic 2 - The Security of Using Large Language Models: A Survey With Emphasis on ChatGPT

Base Paper Details

Authors: Wei Zhou, Xiaogang Zhu, Qing-Long Han, Lin Li, Xiao Chen, Sheng Wen, Yang Xiang

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Publisher: IEEE

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Abstract

The rapid deployment of Large Language Models (LLMs) across multi-domain software interfaces has transformed human-computer interaction, but it also introduces critical system vulnerabilities and security risks. This seminar examines the security landscape of using LLMs, with a particular emphasis on the distinct complexities of conversational models like ChatGPT. The discussion provides a holistic overview of current research categorized into five major threat vectors: structural token bias, automated disinformation generation, behavioral misuse across educational and corporate frameworks, adversarial attacks, and training data privacy violations. The seminar deconstructs the operational mechanics behind prompt injection, universal adversarial triggers (UATs), and hidden backdoor exploits like TROJANLM. Finally, it reviews robust cybersecurity practices and defensive frameworks, including input filtering, role-based access controls, and Selective Differential Privacy (SDP) mechanisms. Based on a comprehensive survey, this seminar establishes a crucial academic foundation for balancing technological utility with responsible, secure AI governance.

Topic 3 - A Survey on Large Language Model based Autonomous Agents

Base Paper Details

Authors: Lei Wang, Chen Ma, Xueyang Feng, Zeyu Zhang, Hao Yang, Jingsen Zhang, Zhi-Yuan Chen, Jiakai Tang, Xu Chen, Yankai Lin, Wayne Xin Zhao, Zhewei Wei, Ji-Rong Wen

Year: 2025

Journal: Frontiers of Computer Science

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Abstract

Autonomous agents have long served as a central pillar in the pursuit of Artificial General Intelligence (AGI), yet traditional approaches are often limited by isolated training environments and simplistic policy functions that diverge from human learning. By utilizing web-scale knowledge, Large Language Models (LLMs) provide a powerful foundation for building human-level autonomous agentic systems. This seminar delivers a systematic review of LLM-based autonomous agents from a unified architectural perspective, focusing on four core functional modules: profiling, memory, planning, and action. The discussion surveys diverse multi-agent applications across social sciences, natural sciences (automated chemical synthesis), and software engineering workflows. Additionally, it explores current objective and subjective evaluation strategies, ranging from human Turing tests to simulated multi-task benchmarking environments. Based on an extensive review, the seminar highlights open research challenges, including prompt robustness, generalized human alignment boundaries, and the transition toward mechanism engineering, providing critical insights into the future of self-driven evolutionary AI.